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Calculation of the Non-Gravitational A_2 Parameter Using Ground-Based Observations of the Apparent Close Approaches between Near-Earth Asteroids and Gaia Stars

M. Yu. Khovrichev[a, b, *] (ORCID: 0000-0001-5910-6713), **D. A. Bikulova**[a] (ORCID: 0000-0001-9394-0816), and **D. L. Gorshanov**[a] (ORCID: 0000-0002-6921-8476)

[a] *The Central Astronomical Observatory of the Russian Academy of Sciences at Pulkovo, St. Petersburg, 196140 Russia*

[b] *Institute of Applied Astronomy, Russian Academy of Sciences, St. Petersburg, 191187 Russia*

**e-mail: gmymax@gmail.com*

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Abstract—The Yarkovsky effect is one of the noticeable factors in the orbital evolution of near-Earth asteroids (NEAs). The A_2 non-gravitational parameter describes the corresponding acceleration in the NEA motion model. This parameter can be derived from astrometric observations of the NEA. We present the results of astrometric observations of two NEAs (2010 XC15 and 2014 HK129). The measurements were performed with the MTM-500M telescope (Mountain Astronomical Station of the Pulkovo Observatory). The modified Gaia star apparent approach technique was applied. As a result, the astrometric accuracy of our observations reached the 0.05 arcsec level. It allowed us to estimate the A_2 values of the 2010 XC15 asteroid: $-139.5 \times 10^{-15} \pm 20.2 \times 10^{-15}$ au/d^2. It is in good agreement with the NASA JPL estimate for this asteroid. The 2014 HK129 A_2 parameter formal value extracted from our data is $61.3 \times 10^{-15} \pm 1583.4 \times 10^{-15}$ au/d^2. Introduction of this A_2 value into the asteroid motion model provides a significant decrease (about 0.05–0.1 arcsec) of the (O–C) values for the first epoch of 2014 HK129 observations. It can be considered as faint evidence of the reality of Yarkovsky drift for the 2014 HK129 asteroid.

Keywords: Yarkovsky drift, appulse, apparent close approach, near-Earth asteroid, A_2 parameter

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INTRODUCTION

The Yarkovsky drift (Yarkovsky, 1901) plays a significant role in the orbital evolution of asteroids. This is concluded in many papers, for example, Michel et al. (2005). As well known, this effect is explained by anisotropic thermal reemission of electromagnetic waves by a small body due to non homogeneous heating and thermophysical properties of different parts of its surface. The value of the effect strongly depends on the rotational state of the body (rotational period and rotational axis orientation). The physical model of the Yarkovsky effect is well-developed as seen from a series of papers (e.g. Vokrouhlicky, 1999; Vokrouhlicky et al., 2000; Martyusheva and Melnikov, 2023). In 1973 Marsden and colleagues (Marsden et al., 1973) proposed to denote the small body nongravitational accelerations as A_1, A_2, A_3. The most significant and potentially detectable part of the Yarkovsky effect is described by the A_2 component, which is pointed perpendicular to the heliocentric radius vector and lies in the current orbital plane. Hence, it leads to a secular change in the orbit's semi-major axis called Yarkovsky drift.

At present time, the A_2 parameter has been estimated for about a few hundred asteroids. First of all, this effect was detected for the near-Earth asteroids due to their relatively small size and suitable observational data coming from ground-based astrometry, radar measurements, and the results of the Gaia space mission. The accuracy of the optical astrometry available from the MPC database is relatively low (about 0.1–0.5 arcsec). Hence, the main expectations and successes are caused by the radar and the Gaia observations (Tanga et al., 2023). The basic strategy for the calculations is varying the state vector for a certain epoch and A_2 parameter according to the observational data fitting procedure (this pipeline is described in the papers Dziadura et al. (2022a, 2022b).

In this study, we performed similar calculations for two NEA (2010 XC15 and 2014 HK129) by adding high-accurate ground-based measurements taken with the Pulkovo Observatory MTM-500M telescope (Gorshanov et al., 2012) to the data set extracted from the MPC database (https://www.minorplanetcenter.net/). We used a technique similar to the observations of apparent approaches (appulses) between Gaia

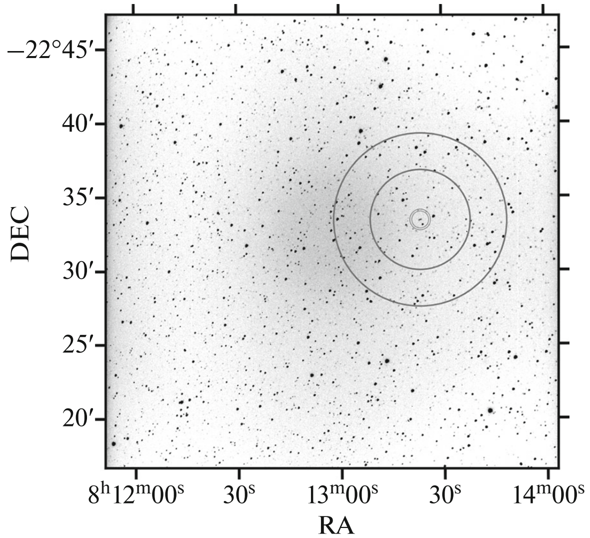

Fig. 1. A typical CCD frame taken with the MTM-500M telescope (exposure time is 20 sec). A small double-circle marks the 2014 HK129 location among stars at 2022-Dec-18 21:14:33.9. The asteroid brightness is 14.5 in V band-pass. An internal asteroid-centered ring is an area for the control stars. The reference stars were taken from the field between the internal (radius is 3 arcmin) and external (radius is 10 arcmin) rings.

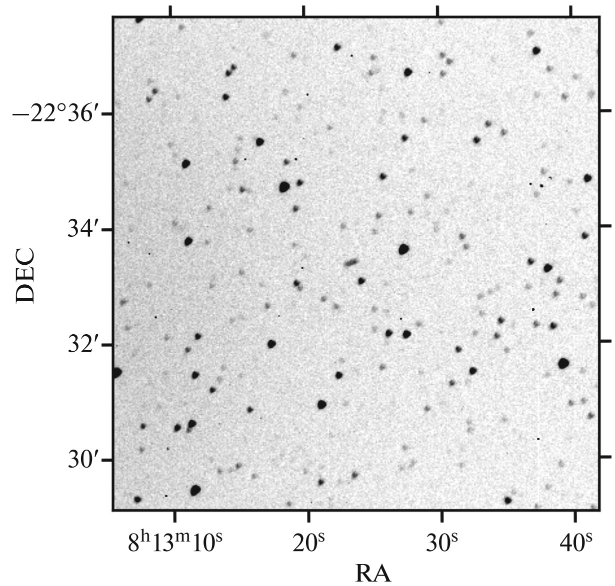

Fig. 2. A small trail at the center of this field is an image of 2014 HK129. This is the magnified part of Fig. 1.

(Gaia Collaboration, 2023) stars and near-Earth asteroids (Bikulova, 2021).

OBSERVATIONS AND DATA ANALYSIS

Observations of "appulses" for two NEA 2010 XC15 (Aten asteroid, $H = 21.48$, MOID = 0.00184 AU) and 2014 HK129 (Apollo asteroid, $H = 21.1$, MOID = 0.00863 AU) were performed with the MTM-500M telescope ($D = 0.5$ m, $F = 4.1$ m, Mountain Astronomical Station of the Pulkovo Observatory, MPC code is C20) in December 2022. This telescope is equipped with the SBIG STX-16803 CCD-camera (4096 × 4096 pixels, the angular scale is 1.34 arcsec/pixel at 3 × 3 pixel binning). Figure 1 shows a typical CCD frame taken with this telescope. The apparent motion rate for both asteroids was in the range of 100 to 2000 arcsec/hr. Hence, the asteroid images were stretched out for the typical exposure times required to collect enough signal value for the background stars (this is seen in Fig. 2).

The astrometric measurements were performed according to the following pipeline. We constructed stellar PSF, reduced to the unit flux, using "good" stars (stars with the mean signal-to-noise ratio). The stellar image fitting was made using the Levenberg–Marquardt technique with fixed PSF by varying photocenter pixel coordinates, background, and flux values. A similar centroiding procedure was applied to the asteroid trail image (the 3D structure of the typical trail is seen in Fig. 3). The image model of the asteroid trail is a convolution of the line segment by the stellar PSF. Hence, we determined the x, y pixel position of the asteroid image barycenter, segment length, positional angle, background level, and flux value. The typical standard errors of the asteroid pixel coordinates was about 0.02 to 0.05 arcsec by taking into account the angular scale of the CCD-camera.

The next stage is astrometric calibration and taking positional systematic errors into account. We divide stars within FOV into two sets. The first of them (a set of control stars) was formed by stars located within the 3 arcmin ring (radius is 3 arcmin). These stars were measured in an "asteroid-mode", i.e. without Gaia usage. The second set includes stars between rings with radii 3 and 10 arcmin. They were used as calibrators to align the pixel coordinates to the Gaia reference frame by the linear reduction model. An example of these areas for both sets of the Gaia stars is presented in Fig. 1. Next, we calculated the equatorial coordinates of the asteroid and control stars and derived the positional corrections to the asteroid coordinates as a mean value of the (O–C) of the control stars.

As a result, we formed the sets of asteroid equatorial coordinates. The set for 2010 XC15 included 288 positions for seven separate series of observations taken during different nights. The 2014 HK129 data set consisted of 80 measurements for three nights.

The usage of all these data for the A_2 estimates may lead to errors because of possible bad measurements in the data set. Therefore, we represented our data by a power series expansion in time and calculated the RA, DEC at the central moment of each CCD frame series. The results are presented in Tables 1 and 2. Here ε_{RA}, ε_{DEC} are internal standard errors of the coordinates. Their values are under 0.05 arcsec. These standard

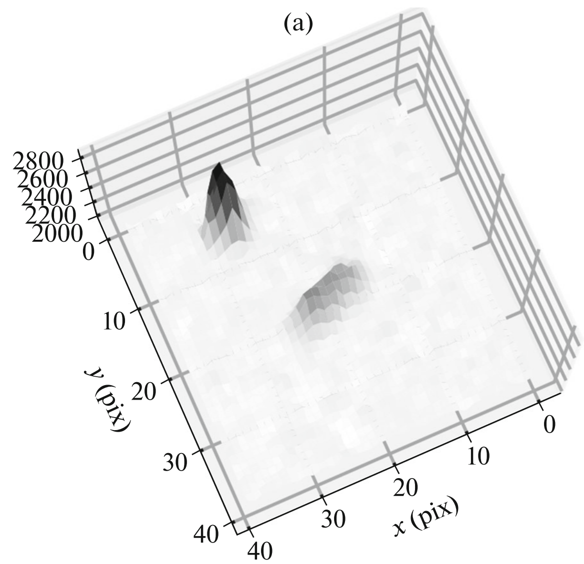

Fig. 3. A 3D structure of the 2014 HK129 image from two various points of view. This plot is based on the same CCD frame as Fig. 2.

errors demonstrate the internal astrometric quality of the measurements. This is clearly shown in Fig. 4. Here the positional residuals as a function of time are plotted. It is seen that the majority of the residuals are in the range of –0.05 to 0.05 arcsec for the longest CCD-series taken for the 2010 XC15. The (O–C) values relative to the NASA JPL Horizons ephemeris are typically less than 0.1 arcsec. The penultimate columns of both tables contain a ratio of the absolute (O–C) value to the standard error formed using ε_{RA}, ε_{DEC}, and the ephemeride standard errors. In most cases, the value of the ratio is within the range of 0.5–1.5, i.e. (O–C) values are in 1-sigma or 2-sigma intervals. This says about the good agreement between observations and ephemerides but doesn't fully exclude the presence of the data that allows us to improve the asteroid ephemerides.

THE A_2 ESTIMATES BASED ON DERIVED COORDINATES AND MPC DATA

The basic condition for the A_2 estimate is the presence of the NEA coordinate measurements related to the several epochs of Earth approaches. This requirement is fulfilled for the 2010 XC15 asteroid. The 2014 HK129 observational history is relatively poor and contains data for two epochs only.

We have mentioned the A_2 calculation pipeline in the introduction. We describe this procedure in more detail here. We extracted the state vector components (position and velocity) at the reference epoch from the SBDB (www.sbdb.jpl.nasa.gov/tools/sbdb_-lookup.html#/) and NASA JPL Horizons system (www.ssd.jpl.nasa.gov/horizons/app.html#/). The asteroid was adopted to be a body of zero mass. The

Table 1. Equatorial coordinates of 2010 XC15 extracted from seven series of CCD frames

JD	RA, deg	DEC, deg	ε_{RA}, arcsec	ε_{DEC}, arcsec	$(O-C)_R$, arcsec	$(O-C)_{DEC}$, arcsec	$(O-C)/\sigma$	V
2459931.441835	142.905258	–2.307441	0.034	0.029	0.029	0.068	1.0	17.4
2459932.399703	143.470556	–2.076437	0.028	0.035	0.020	–0.060	0.8	17.2
2459934.495562	145.128986	–1.302287	0.024	0.011	–0.086	0.119	1.9	16.6
2459934.597768	145.205894	–1.250794	0.021	0.024	0.051	0.093	1.3	16.5
2459937.380670	150.068796	1.228334	0.050	0.025	–0.050	0.051	0.6	15.4
2459938.431686	154.098800	3.383219	0.039	0.024	–0.036	–0.075	0.5	14.8
2459939.385479	161.203603	7.110364	0.039	0.033	–0.027	0.013	0.1	14.2

Table 2. Equatorial coordinates of 2014 HK129 extracted from three series of CCD frames

JD	RA, deg	DEC, deg	ε_{RA}, arcsec	ε_{DEC}, arcsec	$(O-C)_{RA}$, arcsec	$(O-C)_{DEC}$, arcsec	$(O-C)/\sigma$	V
2459930.467814	101.816987	−14.843486	0.029	0.016	−0.080	−0.059	0.7	15.1
2459931.400544	110.120438	−18.196820	0.042	0.028	−0.142	0.037	0.9	14.8
2459932.386826	123.374789	−22.564659	0.063	0.044	−0.102	−0.070	0.6	14.7

positions of the Solar system bodies were taken from the DE441 ephemerides to calculate the gravitational acceleration of the asteroid. The numerical integration of the asteroid motion allowed us to obtain the equatorial coordinates of the asteroid at all epochs of observations and form the corresponding (O–C) values. Afterward, (O–C) was considered as a linear function of the corrections of the state vector components and A_2. We solved the equations using the Levenberg-Marquardt technique. We adopted $A_2 = 0$ at the initial step. As a result, we have improved the asteroid orbit. It means calculating new values of state vector components and A_2 at the reference epoch providing the minimum value of the (O–C) norm.

As we see, one of the results of such a calculation is the A_2 value with its standard error. We followed this idea using the REBOUND package (Rein and Spiegel, 2015) and the ASSIST (Holman et al., 2023) software to perform massless particle motion simulations

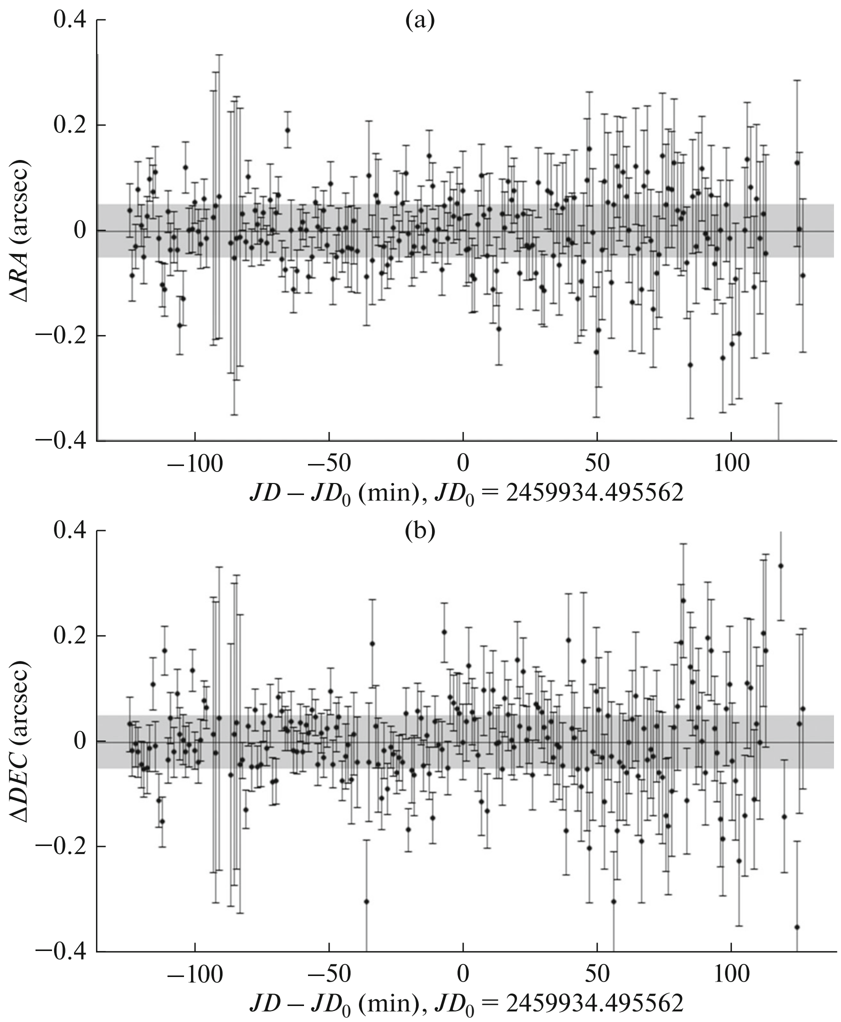

Fig. 4. An example of the residuals vs. time for the 2010 XC15 observations. This is the longest series that consists of 213 measurements.

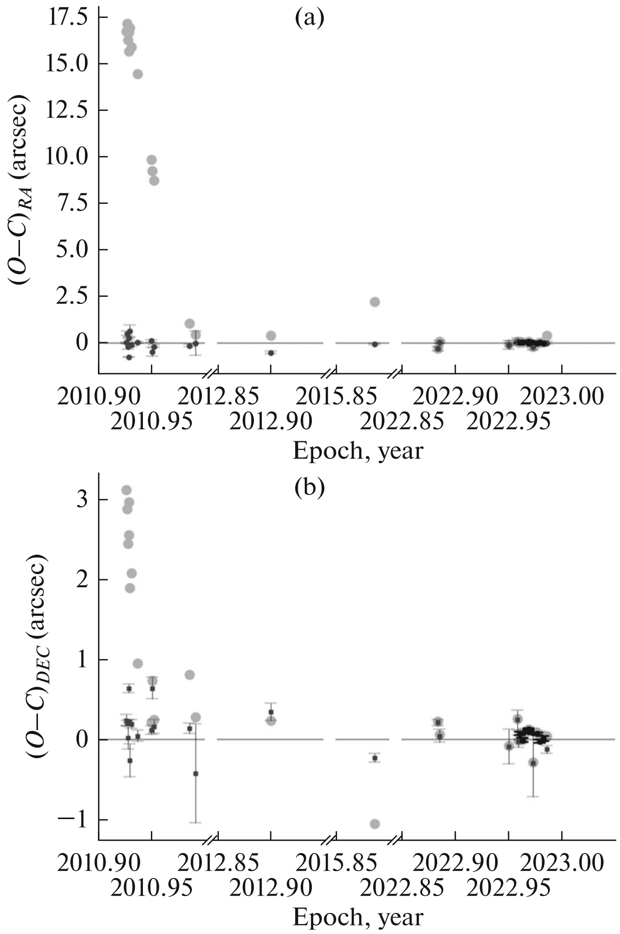

Fig. 5. (O–C) as a function of time for the 2010 XC15 in RA (top panel) and DEC (bottom panel). The light gray points reflect the (O–C) for the NASA JPL state vector and $A_2 = 0$ au/d^2. The points with error bars correspond to our result for $A_2 = -139.5 \times 10^{-15} \pm 20.2 \times 10^{-15}$ au/d^2. Pulkovo data are shown in black.

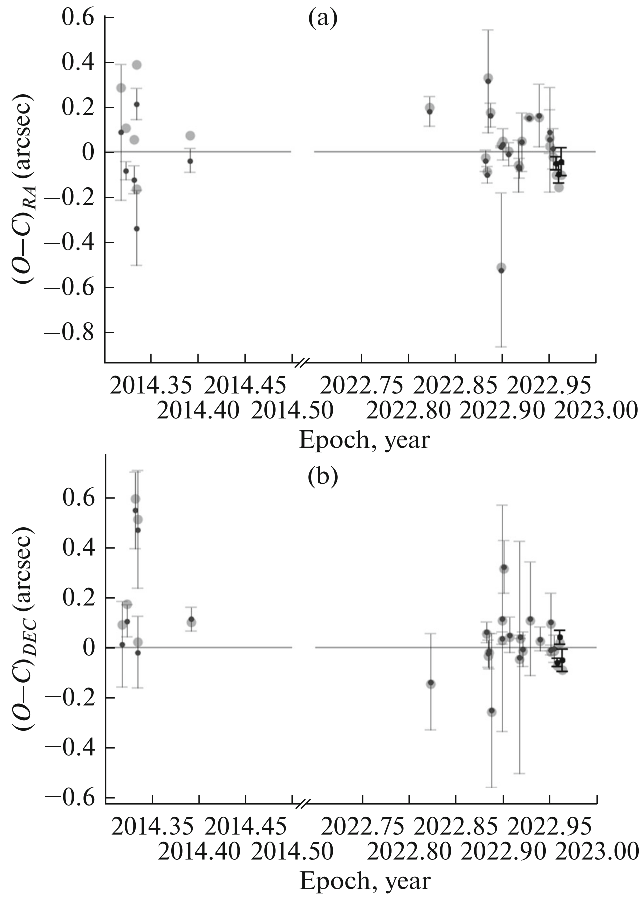

Fig. 6. (O–C) as a function of time for the 2014 HK129 in RA (top panel) and DEC (bottom panel). The light gray points reflect the (O–C) for the NASA JPL state vector and $A_2 = 0$ au/d^2. The points with error bars correspond to our result for $A_2 = 61.3 \times 10^{-15} \pm 1583.4 \times 10^{-15}$ au/d^2. Pulkovo data are shown in black.

in the gravitational field of the Solar system according to NASA JPL DE431 ephemerides (Park et al., 2021).

The (O–C) values vs time for both asteroids are shown in Figs. 5 and 6. The light-gray points demonstrate (O–C) values derived for the case of $A_2 = 0$ au/d^2. The error bars are drawn for the results for which the variation of the A_2 parameter was permitted.

As seen in Figs. 5 and 6, the A_2 fitting procedure should consider the quality of the data provided by different observatories. The natural way is calculating the weights using standard errors of celestial coordinates of the asteroid (ε_{RA}, ε_{DEC}). Unfortunately, the accuracy estimates were not included into the MPC data tables. Therefore, we split them into subsets according to the CCD frames series and similarly performed calculations to the Pulkovo measurement analysis presented in the previous section. The obtained standard error values (ε_{RA}, ε_{DEC}) demonstrate data consistency among each other (internal accuracy). It does not guarantee a good agreement with the external measurement or ephemerides. Hence, a significant absolute value of (O–C)/ε indicates a possible systematic bias for the corresponding data subset (set of CCD frames). We did not have enough information to parametrize the systematic effects. We took it into account by scaling the ε_{RA}, ε_{DEC} by a factor calculating as a mean |(O–C)|/ε for each observational station overall dataset. These scaled ε_{RA}, ε_{DEC} values were applied for the final calculation of the weights. Such an approach is more honest in contrast to $1/\varepsilon^2$ from our point of view. A more detailed discussion on MPC data weighting is present in several papers, for example, the paper published by Farnocchia and colleagues (Farnocchia et al., 2015).

Our final result for 2010 XC15 is $A_2 = -139.5 \times 10^{-15} \pm 20.2 \times 10^{-15}$ au/d^2. The last determination of this value made by NASA JPL and available in the

SBDB database (https://ssd.jpl.nasa.gov/tools/sbdb_lookup.html#/?sstr=2010%20XC15) is $-147.3 \times 10^{-15} \pm 39.3 \times 10^{-15}$ au/d^2. Considering the standard error values, we can conclude an almost complete match between these results. The results of the radar observations were used in NASA JPL calculations. Our results are based on ground-based astrometry data only. Therefore, the natural question is why our observations and analysis are useful. The simplest answer is to exclude the Pulkovo data and recalculate the A_2 value. This manipulation leads to the value $A_2 = -121.9 \pm 30.0 \times 10^{-15}$ au/d^2. It is not dramatically different from the $A_2 = -139.5 \times 10^{-15} \pm 20.2 \times 10^{-15}$ au/d^2 derived using the Pulkovo observations. But the almost 1-σ offset is seen from the NASA JPL result. Hence, our observations and analysis show the significance of the high-accurate ground-based astrometry for the NEA dynamics. We mean that modern radar measurements are available for the relatively small geocentric distances. We have shown that the usage of present time astrometric technique will increase the number of the NEA available for the A_2 estimation.

The 2014 HK129 data set significantly differs from the 2010 XC15 one. It covers a shorter period with fewer data points. The data are separated into two epochs (2014 and 2022). There is a lack of data for the first epoch. The final result for 2014 HK129 is $A_2 = 61.3 \times 10^{-15} \pm 1583.4 \times 10^{-15}$ au/d^2. Hence, we should conclude the statistical insignificance of this estimate. However, we should note a noticeable (O–C) decrease for the first epoch data. Mean (O–C) values of observations made in 2014 were 0.124 and 0.250 arcsec in RA and DEC when $A_2 = 0$ au/d^2. The value of $A_2 = 61.3 \times 10^{-15}$ au/d^2 changes these estimates to -0.048 and 0.205 arcsec in RA and DEC. It is a significant improvement, and the argument supports the reality of the Yarkovsky drift for the 2014 HK129. Indeed, the modern data (2022) are close to the initial state vector epoch (JD = 2460200.5). Hence, it increases the statistical significance of the 2014 observations. The small number of data points and relatively big RMS of (O–C) have led to significant uncertainty in the A_2 estimate for the 2014 HK129.

CONCLUSIONS

The combination of the data from the observation of apparent close approaches between NEA and Gaia stars (the MTM-500M telescope of the Mountain Astronomical Station of the Pulkovo Observatory) and MPC data allows us to perform the A_2 parameter calculation for two near-Earth asteroids. In the case of 2010 XC15, the value $A_2 = -139.5 \times 10^{-15} \pm 20.2 \times 10^{-15}$ au/d^2 is statistically significant. Our 2010 XC15 A_2 value is in good agreement with the previously known value $-147.3 \times 10^{-15} \pm 39.3 \times 10^{-15}$ au/d^2 used in NASA JPL ephemerides.

The mentioned good agreement between A_2 estimates demonstrates the advantage of our astrometric measurements pipeline. As seen from Figs. 5 and 6, the Pulkovo data are characterized by low (O–C) and high accuracy in comparison with other data sources. *Our calculation has shown that excluding the Pulkovo data for the 2010 XC15 leads to a significant systematic offset in the A_2 estimate.*

The A_2 parameter for the 2014 HK129 is equal to $61.3 \times 10^{-15} \pm 1583.4 \times 10^{-15}$ au/d^2, which leads to conclusion about its statistical insignificance. The introduction of this A_2 value into the asteroid motion model provides a significant decrease (about 0.05–0.1 arcsec) of the (O–C) values for the first epoch of observations. It is impossible by orbit improving using $A_2 = 0$ au/d^2. It can be considered as faint evidence of the reality of Yarkovsky drift for the 2014 HK129 asteroid.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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